

HEWANORRA INTL, Saint Lucia

WMO: 789480

Lat: 13.7369N

Lon: 60.9515W

Elev: 33

StdP: 14.68

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00406

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	72.3	73.6	64.3	90.2	77.6	65.9	95.5	78.3	29.5	79.6	27.8	79.7	7.1	40	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	9.4	89.2	79.9	88.2	79.5	88.0	79.5	81.0	86.8	80.5	86.8	80.1	86.2	15.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	151.5	85.2	78.9	150.0	85.2	78.4	147.7	84.8	44.7	87.8	44.3	86.4	43.7	86.6	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	24.5	23.0	68.4	90.6	2.3	1.7	66.8	91.8	65.5	92.8	64.2	93.7	62.5	94.9		
(5)			65.3	84.0	1.7	1.2	64.0	84.9	63.0	85.7	62.0	86.4	60.8	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	79.9	79.7	80.4	81.5	83.0	83.1	83.0	83.2	83.2	82.9	82.1	80.9		
	DBStd	2.06	1.54	1.61	1.76	1.76	1.38	1.36	1.39	1.56	1.73	1.71	1.63	1.57		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11650	926	831	941	944	1023	993	1024	1030	997	1020	962	959		
	CDD65	6175	461	411	476	494	558	543	559	565	547	555	512	494		
	CDH74	67523	4203	3626	4488	5115	6435	6358	6552	6809	6649	6546	5691	5051		
	CDH80	20423	686	597	928	1314	2103	2138	2214	2528	2600	2382	1780	1153		
(14) Wind	WSAvg	15.5	17.1	17.4	16.5	16.2	16.5	18.0	16.8	13.9	11.8	12.6	13.8	16.1		
(15) Precipitation	PrecAvg	63.9	3.7	2.2	2.1	3.7	3.7	6.4	6.8	6.9	7.2	8.7	7.8	4.1		
	PrecMax	84.5	8.3	7.0	6.3	14.7	9.7	15.2	21.2	11.7	18.4	19.3	16.1	8.9		
	PrecMin	45.7	0.7	0.0	0.2	0.6	1.0	2.2	3.3	3.0	1.4	2.2	1.2	0.9		
	PrecStd	12.1	1.7	1.4	1.4	3.2	2.0	3.5	3.5	1.9	3.7	3.9	3.7	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.0	86.2	87.8	88.1	89.4	88.2	88.2	89.7	89.9	89.6	88.2	86.4		
		MCWB	76.2	75.8	77.4	78.5	79.6	79.5	79.6	80.1	80.2	80.1	79.5	77.6		
	2%	DB	84.6	84.6	86.2	87.2	88.1	87.9	87.9	88.2	89.3	88.2	87.7	86.1		
		MCWB	75.5	75.0	76.3	77.9	78.7	79.3	79.5	79.8	80.0	79.7	79.3	77.3		
	5%	DB	84.3	84.3	84.8	86.2	87.5	87.4	87.4	88.0	88.2	87.9	86.5	85.5		
		MCWB	75.3	74.8	75.3	77.1	78.5	78.9	79.2	79.7	79.8	79.7	78.6	76.9		
	10%	DB	83.7	83.4	84.3	85.6	86.3	86.3	86.3	87.5	87.8	87.4	86.1	84.5		
		MCWB	75.0	74.3	74.9	76.7	77.8	78.4	78.6	79.5	79.7	79.4	78.4	76.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.1	77.9	78.7	79.9	80.9	81.0	80.9	81.5	81.7	81.3	80.9	79.7		
		MCDB	83.3	83.9	85.1	86.5	86.9	86.5	86.4	86.8	87.3	86.6	86.3	84.7		
	2%	WB	77.0	76.5	77.4	78.9	79.9	80.1	80.1	80.7	80.8	80.6	80.0	78.5		
		MCDB	83.0	83.0	84.2	85.3	86.2	85.9	86.0	86.9	87.3	87.0	85.8	84.0		
	5%	WB	76.3	75.8	76.5	78.1	79.3	79.7	79.8	80.1	80.1	80.0	79.5	77.9		
		MCDB	82.3	82.3	83.4	84.4	85.5	85.4	85.6	86.2	86.6	86.2	85.3	83.5		
	10%	WB	75.6	75.1	75.8	77.3	78.7	79.1	79.2	79.8	79.8	79.6	79.0	77.2		
		MCDB	81.7	81.6	82.6	83.7	84.9	84.8	84.9	85.7	86.2	85.7	84.7	82.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.1	8.3	8.0	7.8	7.3	7.6	8.5	9.4	9.1	8.7	8.0		
		MCDBR	8.1	8.4	8.5	8.3	8.2	7.7	8.0	8.9	9.7	9.3	8.9	8.5		
	5% WB	MCWBR	4.4	4.5	4.3	4.2	4.1	4.1	4.3	4.5	4.7	4.6	4.7	4.5		
		MCWBR	7.6	8.1	8.1	8.0	8.0	7.5	7.9	8.7	9.4	9.2	8.8	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.369	0.385	0.384	0.425	0.469	0.553	0.513	0.456	0.421	0.383	0.379	0.378		
	taud		2.602	2.533	2.548	2.449	2.305	2.033	2.164	2.386	2.498	2.635	2.630	2.603		
	Ebn at Noon		295	294	295	281	264	240	251	269	280	291	290	290		
	Edh at Noon		30	34	34	37	42	55	48	39	35	30	29	30		
(44) All-Sky Solar Radiation	RadAvg		1661	1874	2009	2058	2006	1911	1953	1982	1916	1738	1598	1578		
	RadStd		56	32	83	82	118	102	83	91	81	74	117	65		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.49	+0.88	N/A	+0.36	N/A	N/A	N/A	N/A	+203	+202

Nomenclature: See separate page